standard. Established by the National Institute of Standards and Technology, it works on the substitution-permutation algorithm of splitting the data in a block size of 128 bits and encrypting it with a 128-bit key length. Major businesses, including banks and financial services, use 128-bit AES data encryption.

Securing device access to the server. To grant end-users application access to the server using their mobile/computer device, the main security measures revolve around user and device authentication. OAuth 2.0 is the most used authorisation and authentication framework. It is being used by firms like Google and Facebook.

MQTT protocol supports the following authentication processes:

1. Username and password or access token. MQTT protocol develops a UTF-8 encoded string user name and a binary data-type password, where each field can be expanded up to 65,535 bytes maximum. The client, i.e., the user, needs to provide a valid username and password to get access from the MQTT broker. Many systems accept passwords instead of tokens, using which the broker can assess the validity, signature and access rights of the user.

2. One-time password (OTP). It is often used for a two-step security, wherein registered devices need to verify their eligibility of access based on an OTP received on the device.

3. Biometrics. This mode of authentication is gaining popularity. Many smartphones and smart devices have introduced fingerprint scanners for access. IoT systems are also incorporating biometric authentication procedure to improve the security measures. Some of the newer bio-authentication processes include facial scan and eye scan.

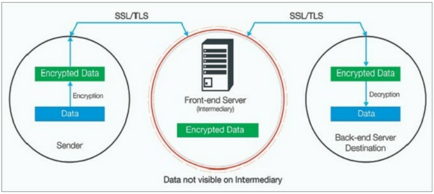

Fig. 3: Golgi IoT end-to-end encryption (Courtesy: www.programmableweb.com)

Hemnani informed, "Like smartphones, IoT devices will also implement fingerprint sensor soon. That way, only an authorised user will be able to access the IoT device."

Security at the edge and component level. Field devices where the data is being generated by the sensors have a temporary storage capacity, which queues the data to be pushed to the server. The major security threats that device storage faces include tampering of the telemetry data, tampering of the cached content and alteration of the device operating system. Simple security options include digital signatures, message authentication code (MAC) and resource access control.

Dr Amarjeet Singh, co-founder and CTO, Zenatix Solutions, explained the process of securing a server from a rogue edge device which might have been physically taken over by a hacker: "A rogue device intruding into an IoT system tries to overload the server by sending an abnormally high amount of data. A security measure taken for server configuration is to limit the amount of data package the server will accept from each device at a given time interval. If any device is transacting higher data than the threshold too often, it is barred from the network." Device gateways that channel sensor data to the server can be secured by BitLocker.

Talking about securing the data in embedded components, Kumar Bhaskar, technology lead of Beyond Evolution, a company working on IoT devices, said, "Components of IoT systems use very limited resources in the form of embedded hardware like a small chip, sensor, controller and RAM. So data is embedded in the controller's memory in encoded or symbolic format. Once the microcontroller is coded, it is locked. So, firstly the data cannot be fetched by any means from the controller. Additionally, the data itself will not be in a readable format."

Software as security suites. Implementing security software has been a staple safety procedure for IoT systems. Here, Sharma shared, "Mostly software solutions are available to secure firmware."

Gartenberg added, "Success of this transformation largely depends on choosing the right IoT software and service provider who can not only catalyse this transformation by understanding business needs but also provide services and software built with security and compliance as major design considerations."

Talking about Microsoft's contribution to IoT security via software, he added, "The Microsoft Azure IoT Suite builds in security measures by design, enabling secure asset monitoring to improve